OPMIG-06: Processing, Parsing & Transformation

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Learning Objectives

By completing this notebook, you will:

1. Master Dynatrace Pattern Language (DPL) for log parsing
2. Configure DQL processors for data transformation

3. ⭐ **NEW:** Parse Apache, Nginx, and JSON logs with production-ready patterns
4. ⭐ **NEW:** Migrate from ELK/Logstash (Grok to DPL conversion)
5. ⭐ **NEW:** Use the complete Parsing Pattern Library (timestamps, stack traces, HTTP)
6. Implement drop processors for cost optimization
7. Validate parsing success rates and troubleshoot failures

Prerequisites

Requirement	Details
Dynatrace Environment	Dynatrace SaaS with Grail and OpenPipeline access — Managed is not covered by this series
Permissions	`settings:read` and `settings:write` (OpenPipeline configuration)
API Access	`logs.read` token scope
DPL Architect	Access to `https://{env}.apps.dynatrace.com/ui/apps/dynatrace.dpl.architect`
Knowledge	OPMIG-01 through OPMIG-05; basic regex or pattern matching familiarity

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Processing Stage Overview

Masking, dropping, parsing and transformation are processors in the **Processing** stage — the first stage of every pipeline. The remaining stages (Smartscape node and edge, Permission, Product and Cost allocation, Bucket assignment, Metric extraction, Davis, Data extraction) follow in a fixed order that you cannot change; OPMIG-02 § *Understanding Processing Order* has the full ten-stage table. This notebook covers the Processing stage; OPMIG-07 covers extraction.

Processor Execution Order

Within the Processing stage, processors execute in the order they're defined, and *"each processor output becomes the input for the next one."* You can reorder them in the UI.

> 💡 **Best Practice:** Order processors logically — mask first, drop second, then parse, then enrich with computed fields based on parsed values.
>
> **Sources:** Processing in OpenPipeline (DT docs) — *"The sequence of stages is fixed for all pipelines and cannot be modified."*

DQL Processor Commands

The DQL processor supports a subset of DQL commands for data transformation.

Available Commands

Command	Purpose	Example
`fieldsAdd`	Add new fields	`fieldsAdd env = "production"`
`fieldsRemove`	Remove fields	`fieldsRemove sensitive_field`
`fieldsRename`	Rename fields	`fieldsRename new_name = old`
`parse`	Extract with DPL	`parse content, "LD? 'count=' INT:count"`
`jsonExtract`	Lift a JSON string's root-level keys to fields	JSON logs
`fieldsFlatten`	Flatten a nested record into fields	after `parse … "JSON:x"`
`fields` / `fieldsKeep`	Keep only the listed fields	

> **Sources:** DQL commands in OpenPipeline (DT docs) — *"jsonExtract Parses a JSON string and extracts the root-level keys of the JSON object to root-level record fields."*

fieldsAdd Examples

Static Values

```
fieldsAdd environment = "production"
| fieldsAdd application = "checkout-service"
| fieldsAdd team = "platform-engineering"
```

Conditional Values (if/else)

```
fieldsAdd severity = if(loglevel == "ERROR", "critical",
                    else: if(loglevel == "WARN", "warning",
                    else: "info"))
```

Computed Values

```
fieldsAdd message_length = stringLength(content)
| fieldsAdd short_host = substring(host.name, from: 0, to: 15)
| fieldsAdd is_error = loglevel == "ERROR"
```

String Operations

```
fieldsAdd normalized_status = lower(status)
| fieldsAdd log_prefix = substring(content, from: 0, to: 50)
| fieldsAdd clean_message = trim(content)
```

Coalesce (First Non-Null)

```
fieldsAdd effective_level = coalesce(loglevel, status, "UNKNOWN")
```

fieldsRemove Examples

```
// Remove a single field
fieldsRemove internal_id
```

```
// Remove multiple fields in one command
fieldsRemove temp_field, debug_info, internal_state
```

fieldsRename Examples

```
// Standardize field names
fieldsRename user_id = userId
| fieldsRename request_id = requestId
| fieldsRename transaction_id = transactionId
```

Dynatrace Pattern Language (DPL)

DPL is a powerful pattern matching language for extracting structured data from text.

Core Matchers

Matcher	Description	Matches
`INT`	Integer	`42`, `-17`, `0`
`LONG`	Long integer	`1234567890123`
`DOUBLE`	Decimal	`3.14`, `-0.5`, `1.0`
`IPADDR`	IP address	`192.168.1.1`, `::1`
`IPV4ADDR`	IPv4 only	`10.0.0.1`
`IPV6ADDR`	IPv6 only	`2001:db8::1`
`LD`	Line data (to delimiter)	Any text until next match
`DATA`	Greedy match	Everything remaining
`SPACE`	Whitespace	Spaces, tabs
`NSPACE`	Non-whitespace	Word-like content
`WORD`	Word chars	`hello`, `user123`
`JSON`	JSON object	`{"key": "value"}`
`EOL`	End of line	Line terminator

Pattern Syntax Elements

Syntax	Meaning	Example
`MATCHER:field`	Extract to named field	`INT:count`
`MATCHER`	Match but don't extract	`SPACE`
`MATCHER?`	Optional match	`(':' INT:port)?`
`'literal'`	Match exact text	`'error_code='`
`(a|b)`	Alternatives	`('user='|'userId=')`
`MATCHER{n,m}`	Bounds the characters the matcher consumes	`WORD{1,3}` on `hello` returns `hel`

> **Quantifier caveat:** `{n,m}` is not universal. `INT` accepts only `*` and `+` — `INT{3}` is rejected. For fixed-width digit runs use a character class: `[0-9]{3}`.

Basic Parse Examples

```
// Extract user ID after "user=", anywhere in the line
parse content, "LD? 'user=' NSPACE:user_id"

// Extract error code (integer), anywhere in the line
parse content, "LD? 'error_code=' INT:error_code"

// Extract IP and port (line starts with the IP)
parse content, "IPADDR:client_ip ':' INT:port"

// Extract JSON payload
parse content, "LD? JSON:payload"
```

> **Three rules that decide whether a pattern returns anything** (each verified with `data record(...)` on 10/02/2026):
> 1. `parse` **anchors at the start of the field.** A pattern that opens with `'user='` returns **null** on every line that does not *begin* with `user=` — no error. Prefix it with `LD?` to skip any leading text (or none): `LD? 'user=' NSPACE:u` on `user=abc x` returns `abc`. At column 0 this can still go wrong: `LD? IPADDR:ip` on a line beginning with the IP returns `92.168.1.100`, and `LD? '"' …` on a line beginning with `"`

> returns null (run 10/30/2026). `parse` does *not* need to consume the whole field; trailing text is ignored.
> 2. **A trailing `LD:x` runs to the end of the line**, not the end of the value — `'requestId=' LD:id` on `requestId=abc-123 status=OK` returns `abc-123 status=OK`. End with `NSPACE:x`, a typed matcher, or a closing literal.
> 3. **`LD` never crosses a line break.** In a multi-line record (a stack trace), `LD? 'at ' …` cannot reach line 2. Use `DATA? 'at ' …` for the first match, or `parseAll(content, "…")` to collect every match as an array.
>
> FAQ-15 covers DPL matching semantics in depth.

Real-World Log Format Examples ⭐ NEW

Production-ready DPL patterns for the most common log formats.

Apache Access Logs (Common Log Format)

Sample: `192.168.1.100 - frank [12/Dec/2024:10:30:45 +0000] "GET /api/users HTTP/1.1" 200 1234`

```
parse content, """
  IPADDR:client_ip SPACE '-' SPACE LD:user SPACE
  '[' TIMESTAMP('dd/MMM/yyyy:HH:mm:ss Z'):log_time ']' SPACE
  '"' LD:method SPACE LD:request_path SPACE LD:protocol '"' SPACE
  INT:status_code SPACE INT:response_bytes
"""
```

Extracted: client_ip, user, log_time, method, request_path, protocol, status_code, response_bytes

Nginx (with Response Time)

Sample: `192.168.1.50 - - [12/Dec/2024:10:30:45 +0000] "POST /api/checkout HTTP/1.1" 201 456 "-" "Mozilla/5.0" "0.342"`

```
parse content, """
  IPADDR:client_ip SPACE '-' SPACE '-' SPACE
  '[' TIMESTAMP('dd/MMM/yyyy:HH:mm:ss Z'):log_time ']' SPACE
  '"' LD:method SPACE LD:request_path SPACE LD:protocol '"' SPACE
  INT:status_code SPACE INT:response_bytes SPACE
  '"' LD:referrer '"' SPACE '"' LD:user_agent '"' SPACE
  '"' DOUBLE:response_time_sec '"'
"""
| fieldsAdd response_time_ms = toLong(response_time_sec * 1000)
```

JSON Application Logs

Sample: `{"timestamp":"2024-12-12T10:30:45.123Z","level":"ERROR","service":"payment-api","message":"Payment gateway timeout"}`

Option 1: **`jsonExtract`** — lifts the root-level keys of the JSON in `content` to fields (OpenPipeline DQL processor command).

Option 2: Parse, then pick or flatten

```
parse content, "JSON:log_data"
| fieldsAdd service = log_data[service]
| fieldsAdd message = log_data[message]
| fieldsAdd level = log_data[level]
```

Syslog (RFC 3164)

Sample: `<34>Dec 12 10:30:45 webserver sshd[1234]: Failed password for admin from 192.168.1.100`

```
parse content, """
  '&lt;' INT:priority '&gt;'
  TIMESTAMP('MMM dd HH:mm:ss'):log_time SPACE
  LD:hostname SPACE LD:app_name '[' INT:pid ']:' SPACE
  DATA:message
"""
// RFC 3164 timestamps carry no year: the parsed value gets the current year
| fieldsAdd facility = toLong(priority / 8)
| fieldsAdd severity = toLong(priority % 8)
```

Java Application Logs (Log4j)

Sample: `2024-12-12 10:30:45,123 [http-nio-8080-exec-5] ERROR com.example.Service - Payment failed`

```
parse content, """
  TIMESTAMP('yyyy-MM-dd HH:mm:ss,SSS'):log_timestamp SPACE
  '[' LD:thread ']' SPACE
  LD:level SPACE LD:logger SPACE '-' SPACE
  DATA:message
"""
```

Stack Trace Extraction:

```
parse content, "LD:exception_class ':' SPACE LD:exception_message EOL"
// Frames are on later lines, and LD stops at a line break — DATA? reaches the first frame
| parse content, "DATA? 'at ' LD:error_location '(' LD:file ':' INT:line_number ')'"
// or collect every frame as an array of records
| fieldsAdd frames = parseAll(content, "LD? 'at ' LD:method '(' LD:file ':' INT:line ')'")
```

Kubernetes / Container Logs

Sample: `2024-12-12T10:30:45.123456789Z stdout F {"level":"info","msg":"Request processed"}`

> The `T` separator is written as a plain letter inside `TIMESTAMP('…')` — a quoted `\'T\'` is rejected (*Named pattern element 'T' is not valid*). The parsed value keeps millisecond precision.

```
parse content, """
  TIMESTAMP('yyyy-MM-ddTHH:mm:ss.SSSSSSSSSZ'):k8s_timestamp SPACE
  LD:stream SPACE LD:log_tag SPACE
  JSON:log_payload
"""
| fieldsAdd level = log_payload[level]
| fieldsAdd msg = log_payload[msg]
```

Common Parsing Patterns

Apache/Nginx Access Logs

Sample log:

```
192.168.1.100 - - [12/Dec/2024:10:30:45 +0000] "GET /api/users HTTP/1.1" 200
1234
```

DPL Pattern:

```
parse content, "IPADDR:client_ip SPACE '-' SPACE LD:user SPACE '[' LD:log_time
']' SPACE '\"' LD:method SPACE LD:path SPACE LD:protocol '\"' SPACE
INT:status_code SPACE INT:bytes"
```

Key-Value Logs

Sample log:

```
userId=12345, action=login, status=success, duration=150ms
```

DPL Patterns:

```
// Extract each key-value pair, wherever it sits in the line
parse content, "LD? 'userId=' INT:user_id"
| parse content, "LD? 'action=' LD:action ','"
| parse content, "LD? 'status=' LD:status ','"
| parse content, "LD? 'duration=' INT:duration_ms 'ms'"
```

Flexible User ID Extraction

Sample logs with varying formats:

```
Processing request for user=john123
User userId=john123 authenticated
Request from user_id=john123 received
```

DPL Pattern (alternatives):

```
parse content, "LD? ('user='|'userId='|'user_id=') NSPACE:user_id"
```

Timestamp Parsing

DPL with timestamp format:

```
parse content, "TIMESTAMP('yyyy-MM-dd HH:mm:ss'):log_timestamp"
parse content, "TIMESTAMP('dd/MMM/yyyy:HH:mm:ss Z'):apache_time"
```

Optional Fields

Sample log:

```
Request to server:8080 completed
Request to server completed
```

DPL Pattern (optional port):

```
parse content, "'Request to ' LD:server (':' INT:port)? ' completed'"
```

Stack Trace Extraction

DPL Pattern:

```
parse content, "LD:exception_class ':' SPACE LD:exception_message"
// first line of the record only — LD stops at the line break
```

Data Transformation Examples

Example 1: Parse and Enrich Application Logs

Input:

```
{"content": "[2024-12-12T10:30:45] ERROR PaymentService - Payment failed for
orderId=12345, amount=99.99"}
```

Processor 1: Parse log structure

```
parse content, "'[' TIMESTAMP('yyyy-MM-ddTHH:mm:ss'):log_time ']' SPACE
LD:level SPACE LD:service ' - ' DATA:message"
```

Processor 2: Extract payment details

```
parse content, "LD? 'orderId=' INT:order_id"
| parse content, "LD? 'amount=' DOUBLE:amount"
```

Processor 3: Add computed fields

```
fieldsAdd severity = if(level == "ERROR", "critical", else: "normal")
| fieldsAdd application_tier = "payment"
```

Example 2: JSON Log Processing

Input:

```
{"content": "{\"level\":\"info\",\"msg\":\"Request
processed\",\"duration_ms\":150}"}
```

Processor: Parse JSON and flatten

```
parse content, "JSON:json_data"
```

To lift the keys to top-level fields, follow with `fieldsFlatten json_data` — or use `jsonExtract` instead of `parse`.

Example 3: Multi-Format Log Normalization

Processor: Normalize different log level formats

```
fieldsAdd normalized_level = if(contains(content, "ERROR") OR
contains(content, "error"), "ERROR",
                            else: if(contains(content, "WARN") OR
contains(content, "warn"), "WARN",
                            else: if(contains(content, "INFO") OR
contains(content, "info"), "INFO",
                            else: if(contains(content, "DEBUG") OR
contains(content, "debug"), "DEBUG",
                            else: "UNKNOWN"))))
```

Parsing Pattern Library ⭐ NEW

Timestamp Patterns (10+ Formats)

Format	Example	DPL
ISO 8601	`2024-12-12T10:30:45Z`	`TIMESTAMP('yyyy-MM-ddTHH:mm:ssZ')`
ISO + MS	`2024-12-12T10:30:45.123Z`	`TIMESTAMP('yyyy-MM-ddTHH:mm:ss.SSSZ')`
Apache	`12/Dec/2024:10:30:45 +0000`	`TIMESTAMP('dd/MMM/yyyy:HH:mm:ss Z')`
Syslog	`Dec 12 10:30:45`	`TIMESTAMP('MMM dd HH:mm:ss')`
Java	`2024-12-12 10:30:45,123`	`TIMESTAMP('yyyy-MM-dd HH:mm:ss,SSS')`
MySQL	`2024-12-12 10:30:45`	`TIMESTAMP('yyyy-MM-dd HH:mm:ss')`
Unix Epoch	`1702380645`	`LONG:epoch` → `toTimestamp(epoch * 1000000000)` (a long is read as nanoseconds)

HTTP Request Patterns

```
// Full request line (a literal " inside the DQL string is escaped as \")
parse content, "LD? '\"' LD:method SPACE LD:path SPACE LD:protocol '\"'"

// Separate path and query
parse content, "LD? '\"' LD:method SPACE LD:path ('?' LD:query)? SPACE
LD:protocol '\"'"

// RESTful API paths (/api/v1/users/12345)
parse content, "LD? '/api/v' INT:api_version '/' LD:resource '/' INT:id"
```

Key-Value Patterns

```
// Simple: user=john count=42
parse content, "'user=' LD:user SPACE 'count=' INT:count"

// Quoted: user="John Doe" email="john@example.com"
parse content, "'user=\"' LD:user '\"' SPACE 'email=\"' LD:email '\"'"

// Logfmt: level=info msg="OK" duration=150ms
parse content, "'level=' LD:level SPACE 'msg=\"' LD:msg '\"' SPACE 'duration='
INT:dur 'ms'"
```

Stack Trace Patterns

```
// Java exception (first line)
parse content, "LD:exception_class ':' SPACE LD:exception_message EOL"

// Stack trace lines — every frame, as an array (LD cannot cross a line break)
fieldsAdd frames = parseAll(content, "LD? 'at ' LD:class_method '(' LD:file
':' INT:line ')'")

// Caused by — anywhere in the record
fieldsAdd causes = parseAll(content, "'Caused by: ' LD:caused_by ':' SPACE
LD:message")

// Python traceback — first frame
parse content, "DATA? 'File \"' LD:file '\", line ' INT:line ', in '
NSPACE:function"
```

PII Masking Patterns

Masking uses `replacePattern`, which takes a **DPL** pattern (not a regular expression — there is no `replaceAll` function). See OPMIG-08 for the full masking set.

```
// Credit cards (4111-1111-1111-1111 → ****-****-****-****); bounded so IDs
and UUIDs are left alone
fieldsAdd content = replacePattern(content, "!&lt;&lt;[0-9A-Za-z-] CREDITCARD
!&gt;&gt;[0-9A-Za-z-]", replacement: "****-****-****-****")

// Email addresses
| fieldsAdd content = replacePattern(content, "[a-zA-Z0-9._%+-]+ '@' [a-zA-Z0-
9.-]+", replacement: "***@***.***")

// SSN (123-45-6789 → ***-**-****)
| fieldsAdd content = replacePattern(content, "[0-9]{3} '-' [0-9]{2} '-' [0-9]
{4}", replacement: "***-**-****")
```

Partial masking ("keep the last 4 digits") needs a lookahead, not a back-reference. `replacePattern` has no back-references, so the old `$4` example cannot be expressed, but a `>>` lookahead checks the last four digits without consuming them, so they survive the replacement. Bound both ends so the pattern cannot fire inside a longer number or an ID:

```
// 4111-1111-1111-1234 → ****-****-****-1234
fieldsAdd content = replacePattern(content, "!&lt;&lt;[0-9A-Za-z-] [0-9]{4} '-
' [0-9]{4} '-' [0-9]{4} '-' &gt;&gt;([0-9]{4} !&gt;&gt;[0-9A-Za-z-])",
replacement: "****-****-****-")
// 4111111111111234 → XXXXXXXXXXXX1234 (unbroken; the letter bound keeps hex
IDs such as 3f2b9c1e4111111111111111aa intact)
| fieldsAdd content = replacePattern(content, "!&lt;&lt;[0-9A-Za-z-] [0-9]{12}
&gt;&gt;([0-9]{4} !&gt;&gt;[0-9A-Za-z-])", replacement: "XXXXXXXXXXXX")
// 123-45-6789 → ***-**-6789
| fieldsAdd content = replacePattern(content, "!&lt;&lt;[0-9A-Za-z-] [0-9]{3}
'-' [0-9]{2} '-' &gt;&gt;([0-9]{4} !&gt;&gt;[0-9A-Za-z-])", replacement: "***-
**-")
```

All three statements were run with `data record(...)` on 10/12/2026 and returned the values in the comments; `card=4111-1111-1111-1111, cvv=123` became `card=****-****-****-1111, cvv=123`.

Network Patterns

```
// IP and port, anywhere in the line
parse content, "LD? IPADDR:ip ':' INT:port"

// IPv4 only
parse content, "LD? IPV4ADDR:ipv4"

// IPv6 only
parse content, "LD? IPV6ADDR:ipv6"

// URL parsing (an LD must be followed by something it can stop at, so the
path starts at '/')
parse content, "LD:protocol '://' LD:hostname (':' INT:port)? '/' LD:path"
```

`LD?` can consume one character before a typed matcher that could also match further in: `LD? IPADDR:ip ':' INT:port` on `192.168.1.100:8080 rest` returns `92.168.1.100`, with no error. If the value can be at column 0, use `IPADDR:ip ':' INT:port` for those lines (run 10/30/2026).

ELK/Logstash Migration Patterns ⭐ NEW

Grok vs. DPL: Key Differences

Grok (Logstash)	DPL (OpenPipeline)
`%{IP:client_ip}`	`IPADDR:client_ip`
`%{INT:count}`	`INT:count`
`%{WORD:user}`	`WORD:user`
`%{GREEDYDATA:msg}`	`DATA:msg`
`\\s+`	`SPACE`

Apache Log Migration

Grok: `%{COMBINEDAPACHELOG}`

DPL:

```
parse content, """
  IPADDR:client_ip SPACE LD:ident SPACE LD:auth SPACE
  '[' TIMESTAMP('dd/MMM/yyyy:HH:mm:ss Z'):log_time ']' SPACE
  '"' LD:method SPACE LD:request_path (SPACE LD:http_version)? '"' SPACE
  INT:status_code SPACE (INT:response_bytes | '-') SPACE
  '"' LD:referrer '"' SPACE '"' LD:user_agent '"'
"""
```

Logstash Filter → OpenPipeline Mapping

Logstash Filter	OpenPipeline
`grok { }`	DQL parse processor
`mutate { add_field }`	`fieldsAdd field = "value"`
`mutate { remove_field }`	`fieldsRemove field`
`mutate { rename }`	`fieldsRename new = old`
`drop { }`	Drop processor
`json { }`	`jsonExtract`, or `parse content, "JSON:x"` + `fieldsFlatten x`
`date { }`	TIMESTAMP in parse

Complete Migration Example

Logstash:

```
filter {
  if [level] == "DEBUG" { drop { } }
  grok { match =&gt; { "message" =&gt; "%{TIMESTAMP_ISO8601:ts} %
{LOGLEVEL:level} %{GREEDYDATA:msg}" } }
  mutate { add_field =&gt; { "env" =&gt; "production" } }
}
```

OpenPipeline:

1. Drop processor: `loglevel == "DEBUG"`
2. DQL parse: `parse content, "TIMESTAMP('yyyy-MM-ddTHH:mm:ss.SSSZ'):ts SPACE`

`LD:level SPACE DATA:msg"`
3. DQL add: `fieldsAdd env = "production"`

Technology Bundle Parsers

The **Technology bundle** processor applies a library of predefined processors for a selected technology. The documented library is grouped by category, for example:

Category	Examples from the library
Servers	Apache HTTP (access, error logs), Nginx (access, error log), Apache Tomcat, HAProxy
Syslog	RFC 3164, RFC 5424, Generic, Cisco switch
Applications	Java (Spring Boot, Slf4j, Log4j, Logback, Util Logging, exception stack trace), Node.js, Go
AWS / Azure	API Gateway, CloudFront, EKS control plane, AKS, and more

There is no generic "JSON" bundle — for JSON logs use `jsonExtract` or `parse … "JSON:x"` in a DQL processor. The fields a bundle adds are defined by the bundle; check its output on a sample record before you rely on a field name.

> **Sources:** Technology bundle processors for logs (DT docs) — *"The following table lists the technology bundles available in OpenPipeline."* (library read 10/02/2026).

When to Use Technology Parsers

Use Technology Parser	Use Custom DQL/DPL
Standard log format	Custom/proprietary format
Quick setup needed	Specific field extraction
Common technology	Conditional parsing

Use Technology Parser	Use Custom DQL/DPL
All fields needed	Selective extraction

Configuring Technology Parsers

1. Open pipeline in OpenPipeline settings
2. Go to the **Processing** stage
3. Click **+ Processor** → **Technology bundle**
4. Select the technology
5. Configure matching condition
6. Save

Drop Processors

Drop processors remove records from the pipeline before storage.

Common Drop Patterns

Use Case	Matching Condition
Debug logs	`loglevel == "DEBUG"`
Trace logs	`loglevel == "TRACE"`
Health checks	`matchesValue(content, "*health*")`
Readiness probes	`matchesValue(content, "*/ready*")`
Metrics endpoints	`matchesValue(content, "*/metrics*")`
Heartbeats	`matchesPhrase(content, "heartbeat")`
Specific source	`log.source == "noisy-service"`

> Matching conditions accept `matchesValue` (with `*` wildcards, any substring) and `matchesPhrase` (whole tokens); `contains()` is not enabled in a matcher.

Drop Processor Configuration

```
Processor Type: Drop
Name: Drop debug logs
Matching Condition: loglevel == "DEBUG"
```

Combining Drop Conditions

```
// Drop all non-essential logs
loglevel == "DEBUG"
  OR loglevel == "TRACE"
  OR matchesValue(content, "*health*")
  OR matchesValue(content, "*/metrics*")
```

> ⚠️ **Important:** Dropped data is gone forever. Test drop conditions carefully before deploying.

Advanced Processing Patterns

Pattern 1: Fix Missing Timestamp

When logs have a custom timestamp format that's not recognized:

```
// Parse timestamp from content
parse content, "'[' TIMESTAMP('yyyy-MM-dd HH:mm:ss'):parsed_timestamp ']'"
```

Parse into a **new** field (`parsed_timestamp`, `log_time`). Overwriting `timestamp` risks the record failing log schema validation — a stored log must have a `timestamp` *"Within the ingestion range"*, and one that is not is dropped. Set `timestamp` deliberately and only from a parsed `TIMESTAMP` value.

> **Sources:** OpenPipeline limits (DT docs) — *Schema validation for logs*: *"A processed log record is persisted only if all the following field conditions are satisfied. If the schema is not valid, the log is*

> *dropped."*

Pattern 2: Fix Missing Log Level

When logs don't have a standard loglevel field:

```
// Extract level from content
fieldsAdd loglevel = if(contains(content, "[ERROR]") OR contains(content,
"ERROR:"), "ERROR",
                    else: if(contains(content, "[WARN]") OR contains(content,
"WARN:"), "WARN",
                    else: if(contains(content, "[INFO]") OR contains(content,
"INFO:"), "INFO",
                    else: if(contains(content, "[DEBUG]") OR
contains(content, "DEBUG:"), "DEBUG",
                    else: "NONE"))))
```

Pattern 3: Parse JSON from Within Text

When JSON is embedded in a log message:

```
// Extract JSON from text (trailing text after the JSON is ignored)
parse content, "LD? JSON:embedded_json"
```

Pattern 4: Compute Response Time Categories

```
fieldsAdd response_category = if(duration_ms &lt; 100, "fast",
                              else: if(duration_ms &lt; 500, "normal",
                              else: if(duration_ms &lt; 1000, "slow",
                              else: "very_slow")))
```

Pattern 5: Standardize Boolean Fields

```
fieldsAdd is_success = if(status == "success" OR status == "ok" OR status ==
"200", true, else: false)
```

Pattern 6: Extract Service Name from Path

```
// From "GET /api/v1/users HTTP/1.1" → users
parse content, "LD? '/api/v' INT '/' NSPACE:service_name"
```

Validating Your Processing

After configuring processors, validate that parsing is working correctly.

```
// Check parsing success rate across all pipelines
// loglevel and status are never null: a record with no detected level reads
NONE
fetch logs, from: now() - 1h
| summarize {
    total = count(),
    with_loglevel = countIf(loglevel != "NONE")
  }
| fieldsAdd parsing_rate = round((toDouble(with_loglevel) / toDouble(total)) *
100, decimals: 1)
```

```
// Parsing success rate by pipeline
fetch logs, from: now() - 1h
| filter isNotNull(dt.openpipeline.pipelines)
| summarize {
    total = count(),
    parsed = countIf(loglevel != "NONE")
  }, by: {dt.openpipeline.pipelines}
| fieldsAdd parsing_rate = round((toDouble(parsed) / toDouble(total)) * 100,
decimals: 1)
| sort parsing_rate asc
```

```
// Sample logs that failed parsing (no level detected: loglevel is NONE)
fetch logs, from: now() - 1h
| filter loglevel == "NONE"
| fields timestamp, log.source, dt.openpipeline.pipelines, content
| limit 25
```

```
// Verify specific parsed fields exist
// Replace 'your_field' with fields your parsing should create
fetch logs, from: now() - 1h
| filter isNotNull(dt.openpipeline.pipelines)
| summarize {
    total = count(),
    with_user_id = countIf(isNotNull(user_id)),
    with_request_id = countIf(isNotNull(request_id))
  }, by: {dt.openpipeline.pipelines}
```

```
// Sample successfully parsed logs to verify field extraction
fetch logs, from: now() - 1h
| filter loglevel != "NONE"
| limit 20
```

```
// Verify drop processors are working
// If drops are configured, debug log count should be low
fetch logs, from: now() - 1h
| summarize {debug_count = countIf(loglevel == "DEBUG")}
| fieldsAdd message = if(debug_count == 0, "✅ Drop processor working - no
debug logs",
                       else: "⚠️ Debug logs still present - check drop
configuration")
```

```
// Check log level distribution after processing
fetch logs, from: now() - 1h
| summarize {log_count = count()}, by: {loglevel}
| sort log_count desc
```

DPL Architect Tool

Dynatrace provides a **DPL Architect** tool for building and testing patterns:

Accessing DPL Architect

1. Navigate to **Settings** → **OpenPipeline**
2. When adding a parse processor, click **Open DPL Architect**

3. Or access via: `https://{your-environment}.apps.dynatrace.com/ui/apps/dynatrace.dpl.architect`

Using DPL Architect

1. Paste sample log content
2. Build pattern interactively
3. See extracted fields in real-time
4. Copy pattern to processor definition

> 💡 **Tip:** Always test patterns in DPL Architect before deploying to production pipelines.

Complete Processing Pipeline Example

Pipeline: `application-logs`

Processor 1: Drop Debug (Drop)

```
Matching: loglevel == "DEBUG" OR matchesValue(content, "*[DEBUG]*")
```

Processor 2: Parse Application Log (DQL)

```
parse content, "'[' TIMESTAMP('yyyy-MM-dd HH:mm:ss'):log_ts ']' SPACE '['
LD:level ']' SPACE '[' LD:thread ']' SPACE LD:class ' - ' DATA:message"
```

Processor 3: Extract Request ID (DQL)

```
parse content, "LD? 'requestId=' NSPACE:request_id"
```

Processor 4: Add Environment Tags (DQL)

```
fieldsAdd environment = "production"
| fieldsAdd application = "checkout-service"
| fieldsAdd team = "platform"
```

Processor 5: Compute Severity (DQL)

```
fieldsAdd severity = if(level == "ERROR", "P1",
                     else: if(level == "WARN", "P2",
                     else: "P3"))
```

Next Steps

Now that you can transform data, continue with:

Notebook	Focus Area
OPMIG-07	Metric & Event Extraction
OPMIG-08	Security, Masking & Compliance
OPMIG-09	Troubleshooting & Validation

References

- OpenPipeline Processing
- Processing Examples
- Dynatrace Pattern Language
- DPL Architect Tool
- DQL Functions in OpenPipeline

This notebook was AI-generated from community-submitted and publicly available sources. This notebook series is not officially supported by Dynatrace. Always verify information against official Dynatrace documentation.